Real analysis

1. Sets, relations, functions, and order

1.1. Logic

Theorem 1.1.1 (De Morgan's Laws for statements)

AND and OR interchange under complementation.

In C notation:

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!(A || B) == (!A) && (!B)
!(A && B) == (!A) || (!B)
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In mathematical logical notation:

$$\neg(A \vee B) = (\neg A) \wedge (\neg B)$$
$$\neg(A \wedge B) = (\neg A) \vee (\neg B)$$

Theorem 1.1.2 (De Morgan's Laws for set theory)

Union and intersection interchange under complementation.

$$(A \cup B)^c = A^c \cap B^c$$
$$(A \cap B)^c = A^c \cup B^c$$

Theorem 1.1.3

Given the true statement $P \Longrightarrow Q$,

- $\neg P \Longrightarrow \neg Q$ is the ***inverse***, which might be true or false.
- $Q \Longrightarrow P$ is the ***converse***, which is true iff the inverse is true.
- $\neg(P \Longrightarrow Q)$ is the ***negation***, which is always false.
- $\neg Q \Longrightarrow \neg P$ is the ***contrapositive***, which is always true.

1.2. Sets

See 04 Set Theory, [Topology] 01 Set Theory

A set is defined by selecting elements from another set such that $S(x)$ is true:

$$\{x \in A : S(x)\} \text{ or } \{x \in A \mid S(x)\}$$

$\varnothing$ denotes the empty set.

$P \Rightarrow Q$ means P implies Q (if P, then Q).

$P \Leftrightarrow Q$ means P and Q are equivalent (P iff Q).

$A \subset B$ means A is a subset of B (**$A = B$ is allowed**).

$A \subsetneq B$ means A is a strict/proper subset of B.

$A = B$ means $A \subset B$ and $B \supset A$.

$A \cup B$ means the union of A and B.

$A \cap B$ means the intersection of A and B.

$A \parallel B$ means $A \cap B = \varnothing$.

$A \setminus B = \{x \in A : x \notin B\}$, everything in A that's not in B.

$A \times B = \{(a, b) : a \in A, b \in B\}$ is called the ***Cartesian product*** of A and B.

Definition 1.2.1 (Power set)

The set of all subsets of A, denoted $\mathcal{P}(A)$.

1.3. Ordered pair

See 04 Set Theory

Definition 1.3.1 (Ordered pair)

The ***ordered pair*** (a, b) is the set $\{\{a\}, \{a, b\}\}$.

Theorem 1.3.2

$(a, b) = (c, d) \Longleftrightarrow a = c$ and $b = d$.

1.4. Relations

See 04 Set Theory, [Topology] 02 Relations

Definition 1.4.1 (Relation)

A ***relation*** $R : A \to B$ is any subset of $A \times B$.

aRb means $(a, b) \in R$.

Theorem 1.4.2 (Relations can compose)

If $R : A \to B$ and $S : B \to C$, then $S \circ R : A \to C$ is a subset of $A \times C$.

⚠ **Warning**

$R \circ R^{-1}$ is not necessarily the identity map!

Theorem 1.4.3

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}$$
$$(T \circ S) \circ R = T \circ (S \circ R)$$

Definition 1.4.4

R is ***reflexive*** iff aRa for all $a \in A$.

R is ***symmetric*** iff $aRb \Leftrightarrow bRa$.

R is ***transitive*** iff $aRb, bRc \Rightarrow aRc$.

1.4.1. Equivalence relations

Definition 1.4.5 (Equivalence relation)

A relation that is reflexive, symmetric, and transitive.

Definition 1.4.6 (Equivalence class)

Let $R : A \to B$ be an equivalence relation. Then the ***equivalence class*** of $a \in A$ is

$$[a] = [a]_R := \{b \in A : aRb\}$$

The set of all equivalence relations of R in A is called A/R.

Lemma 1.4.7

Two equivalence classes are either identical or disjoint.

Definition 1.4.8 (Partition)

A collection of disjoint nonempty subsets of A whose union is all of A.

Theorem 1.4.9

Given an equivalence relation on nonempty A, its set of equivalence classes forms a partition of A.

Given a partition P of A, there is a unique equivalence relation whose set of equivalence classes forms P.

1.4.2. Functions (maps)

See 03 Functions

Definition 1.4.10 (Function)

A ***function*** (also called a ***map*** or ***mapping***) is a relation $f : A \to B$ where each $a \in A$ appears as the first element (input) of an ordered pair $(a, b) = afb$ exactly once.

If $(a, b) \in f$, we can also say $f(a) = b$. b is the ***value*** of f at a, or the ***image*** of a under f.

A is the ***domain*** and B the ***codomain*** of f.

Definition 1.4.11 (Injective)

A function f is ***injective*** or ***one-to-one*** iff:
- $f(a) = f(b) \Rightarrow a = b$, or equivalently
- $a \neq b \Rightarrow f(a) \neq f(b)$

Definition 1.4.12 (Surjective)

A function $f : A \to B$ is ***surjective*** or ***onto*** iff:
- $f(A) = B$ (its image is its codomain), or equivalently
- for every $b \in B$ there exists $a \in A$ s.t. $f(a) = b$

Definition 1.4.13 (Bijective)

A function is ***bijective*** and has ***one-to-one correspondence*** if it is both injective and surjective.

Theorem 1.4.14

A composition of two functions is a function. If both are sur/in/bijective, so is their composition.

Definition 1.4.15

The set of all functions $A \to B$ is denoted B^A.

Lemma 1.4.16

If A and B are finite, then the size of B^A is $|B|^{|A|}$, the size of B to the size of A.

Theorem 1.4.17

For any set A, there is a bijection between $\mathcal{P}(A)$ and 2^A.

Definition 1.4.18 (Induced map)

The ***induced map*** of the relation $f : A \to B$ is a function $\overline{f} : \mathcal{P}(A) \to \mathcal{P}(B)$ defined as

$$\overline{f}(X) := \{b \in B : b = f(a) \text{ for some } x \in A\}$$

$\overline{f}(X)$ is called the ***image*** of X. This is sometimes denoted $f(X)$ (which is an abuse of notation).

$\overline{f}$ is always a function regardless of if f is a function.

1.5. Sequence

Definition 1.5.1 (Sequence)

A ***sequence*** is a function whose domain is the natural numbers, or some subset of the natural numbers.

If a is a sequence, $a(n)$ is also denoted a_n, and is called the nth element of a.

1.6. Cardinality

See 06 Counting and Sizes of Sets

Assume that there is some "overset" containing all of the sets we consider.

Definition 1.6.1 (Cardinality)

$\operatorname{card}(A)$ is the equivalence class of A under bijection.

Equivalently, $\operatorname{card}(A) = \operatorname{card}(B)$ iff there is a bijection $A \to B$.

Two sets are ***equipotent***, ***equinumerous***, ***equipollent***, or ***similar*** if they have the same cardinality.

Definition 1.6.2

$\operatorname{card} A \leq \operatorname{card} B$ iff there is an injection from A to B.

Theorem 1.6.3 (Schroeder-Bernstein Theorem)

$$\operatorname{card}(A) \leq \operatorname{card} B \text{ AND } \operatorname{card}(B) \leq \operatorname{card}(A) \\ \Longrightarrow \operatorname{card}(A) = \operatorname{card}(B)$$

Theorem 1.6.4 (Cantor's Theorem)

For any set A,

$$\operatorname{card}(A) < \operatorname{card}(\mathcal{P}(A))$$

Theorem 1.6.5

Every infinite set is similar to a proper subset of itself.

No finite set is similar to a proper subset of itself.

1.7. Countability

Definition 1.7.1 (Countability)

S is countable iff there is an injection $f : S \to \mathbb{Z}^+$, or equivalently iff card $S \leq$ card $\mathbb{Z}^+$.

Corollary 1.7.1.1

1. Every subset of a countable set is countable.
   - The elements of a countable set can be indexed by the positive integers.
   - All finite sets are countable.
2. A countable union of countable sets is countable.

Theorem 1.7.2

The product of a finite number of countable sets is countable.

However, a product of an infinite number of countable sets is not countable.

Theorem 1.7.3

$$\operatorname{card}(\mathbb{R}) = \operatorname{card}(\mathcal{P}(\mathbb{Z}^+))$$

Thus, the reals are not countable.

1.8. Axiom of choice

Axiom 1.8.1 (Axiom of Choice)

Given a infinite collection of disjoint finite sets, it is possible to select one element from each of the sets.

Equivalently, if A is a set of nomempty sets, there is always a function $f : A \to \cup A$ such that $f(a) \in a$ for all $a \in A$.

In this class, we assume the Axiom of Choice.

Theorem 1.8.2 (Zermelo's Theorem)

The following theorems are equivalent:
- the Axiom of Choice
- the Well-Ordering Property
- the Hausdorff Maximum Principle
- Zorn's Lemma

Theorem 1.8.3 (Well-Ordering Property (Zermelo's Postulate))

Every set can be well-ordered.

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Theorem 1.8.4 (Hausdorff Maximum Principle)

Every poset has a "maximal" totally ordered subset, i.e. a totally ordered subset that cannot be extended without escaping the poset.

Theorem 1.8.5 (Zorn's Lemma)

If A is a poset in which every totally ordered subset has an upper bound, then A has a maximal element, i.e. there is no larger element in the poset.

2. Fields and order

2.1. Equality

Definition 2.1.1 (Equality)

Equality is an equivalence relation on the reals, so:
- **Reflexive:** $x = x$.
- **Symmetric:** $x = y \Longrightarrow y = x$.
- **Transitive:** $x = y$ and $y = z \Longrightarrow y = z$.

2.2. Fields

Definition 2.2.1 (Field)

A ***field*** $\mathbb{F}$ is any set equipped with two relations, ***addition*** $(+)$ and ***multiplication*** $(*)$, which satisfy the following properties:
- **Closure axiom:** If x and y are in $\mathbb{F}$, so are both $x + y$ and xy.
- **Commutativity axioms:** $x + y = y + x$ and $xy = yx$.
- **Associativity axioms:** $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$.
- **Distributivity axiom:** $x(y + z) = xy + xz$.
- **Substitution axioms:** $w = x$ and $y + z \Longrightarrow w + y = x + z$ and $wy = xz$.
- The **additive inverse axiom** and **multiplicative inverse axiom**, defined below.

Axiom 2.2.2 (Additive inverse axiom)

Given $x, y \in \mathbb{F}$, there exists $z \in \mathbb{F}$ s.t. $x + z = y$ (i.e. all addition problems have solutions).

We denote this z by the expression $(y - x)$. We also denote, for every x:

$$0_x := (x - x)$$
$$-x := (0_x - x)$$

Theorem 2.2.3 (Zero is unique)

If $\mathbb{F}$ satisfies the additive inverse property, there is some $0 \in \mathbb{F}$ s.t. $0 = 0_x = 0_y$ for all x and y in $\mathbb{F}$.

Axiom 2.2.4 (Multiplicative inverse axiom)

There exists some number in $\mathbb{F}$ which is not equal to 0. If $x, y \in \mathbb{F}$ s.t. $x \neq 0$,

3. Uncategorized

Lemma 3.1

The positive integers have no upper bound.

Proof. Assume that $z = \sup \mathbb{Z}^+$ exists. We know that $z - 1 < z$, so by the approximation property there is a positive integer s with $z - 1 < s$. Thus $z < s + 1$. □

Definition 3.2 (Comensural)

The set X is ***comensural*** iff for any $x, y \in X$ with $x > 0$, then there is a positive integer n s.t. $nx > y$.

Theorem 3.3 (Archimedean property)

The reals are comensural.

Proof. Consider $\frac{y}{x}$. If $\frac{y}{x} < 0$, then $n = 0$. Otherwise, $\frac{y}{x} > 0$. Since the positive integers are unbounded, choose $n > \frac{y}{x}$. Then $nx > y$. □

Theorem 3.4 (Density of the reals / rationals between reals)

Between any two reals there is a rational number.

Proof. The other cases are trivial, so consider $0 < x < y$. Choose an integer $n > \frac{1}{y-x}$.

We define $q := \frac{k}{n}$, and

We claim $x < q$. Then:

$$\frac{1}{y - x} < n$$

$$y - x > \frac{1}{n}$$

$$y - \frac{1}{n} > x$$

$$y - q \leq \frac{1}{n} < y - x$$

□

3.1. Extended reals

Definition 3.1.1 (Extended reals)

The reals, plus two elements $-\infty$ and $+\infty$.

The order on it is defined such that

$$-\infty < x < +\infty \text{ for all } x \in \mathbb{R}$$

and when comparing two reals it still has the standard order.

All expressions whose value is defined follow the rules of complete ordered fields, but there are many undefined expressions.

3.2. Absolute values

Theorem 3.2.1 (Corollaries of the Triangle Inequality)

For any $x, y \in \mathbb{R}$:

$$|x + y| \leq |x| + |y|$$
$$|x - y| \leq |x| + |y|$$
$$||x| - |y|| \leq |x - y|$$

3.3. Boundedness

Definition 3.3.1 (Boundedness)

$f : S \to \mathbb{R}$ is ***bounded*** iff the image $f(S)$ has both a supremum and infimum.

Equivalently, f is bounded iff there exists M s.t. $|f(x)| < M$ for all x.

Equivalently, f is bounded iff $f(S)$ is bounded.

Theorem 3.3.2

If $f(x) \leq g(x)$ for all $x \in D$, then $\sup f(D) \leq \sup g(D)$.

3.4. Decimal representation

Definition 3.4.1 (Decimal representation)

A string of the form

$$(-)d_n d_{n-1} ... d_1 d_0 . d_{-1} d_{-2} ...$$

where each $d_i \in \{0, 1, 2, ..., 9\}$.

i.e.

- optionally, a minus sign
- a finite string of digits
- a decimal point
- an infinite string of digits

Theorem 3.4.2

There is a bijection between $\mathbb{R}^+$ and the set of decimal representations without a minus sign and such that there is a last 9, i.e. the set of nonterminating positive decimal representations.

In order to prove this, we need some lemmas:

Lemma 3.4.3

There is a bijection between $\mathbb{Z}^+$ and the set of decimal representations ending in an infinite string of 0s.

Proof. Every integer is representable as

$$n = 10m + r.$$

So keep dividing the integer by 10, and pushing the remainder onto the front of the string. This gives us a base-10 representation of the integer, i.e. a finite string of digits.

To make a full decimal representation, put an infinite string of 0s after the decimal point. □

Now let's prove the original theorem.

Proof. Let $x \in \mathbb{R}^+$. Take $n = \lfloor x \rfloor$ and find its decimal representation. Then $0 < x - n < 1$. □

Now we need to prove more lemmas:

Lemma 3.4.4

Every decimal representation represents a real number.

4. Sequences

4.1. Subsequences

Definition 4.1.1 (Subsequence)

If $f : \mathbb{Z}^+ \to \mathbb{Z}^+$ is a strictly increasing function

4.2. Limits

TODO

Theorem 4.2.1 (Squeeze Theorem)

Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be sequences s.t. $a_n \le b_n \le c_n$ for all n and $a_n \to L$ and $c_n \to L$. Then $b_n \to L$.

Proof. Let $\varepsilon > 0$. Choose n so large that both $|a_n - L|, |c_n - L| < \varepsilon$. Then,

$$-\varepsilon < c_n - L < \varepsilon$$
$$-\varepsilon < a_n - L < \varepsilon$$

and thus

$$-\varepsilon < a_n - L \le b_n - L \le c_n - L < \varepsilon$$

so clearly

$$|b_n - L| < \varepsilon.$$

Since this works for all ε, $b_n \to L$. □

Theorem 4.2.2 (Ratio Test)

Proof. Casework!

$L < 1$: Choose r s.t. $L < r < 1$. For any $\varepsilon \in (0, r - L)$, find n s.t.

$$\frac{|x_n + 1|}{|x_n|} < r.$$

Then, since $\varepsilon < r - L$, subtracting L from both sides gives us □

In the book, it proves

Theorem 4.2.3

$$\lim_{n\to\infty} n^{\frac{1}{n}} = 1$$

Proof. $n^{\frac{1}{n}}$ is definitely bigger than 1, since otherwise n would be a power of something less than 1.

If you have $(1+\varepsilon)^n$, it happens to be less than $1 + n\varepsilon$.

Thus,

$$1 + \varepsilon < (1 + n\varepsilon)^{\frac{1}{n}}.$$

So, we'd like to show that

$$\lim_{n\to\infty} \frac{n}{(1+\varepsilon)^n} = 0.$$

□

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